

HalluciGuard Detector Agent

System Architecture & Technical Specification | Version 2.0.0

1. Executive System Overview

HalluciGuard is a multi-signal agentic AI system designed to detect and quantify hallucinations in Large Language Model (LLM) outputs. The **Detector Agent** synthesizes four statistical uncertainty signals: Token Probability, Predictive Entropy, Semantic Similarity, and conditionally gated Self-Consistency. An **Intelligent Gating Mechanism (SelfConsistencyGate)** triggers multi-sample stochastic decoding only when single-pass uncertainty evaluates to MEDIUM risk on analytical query categories, reducing compute overhead by over 75%.

2. Detection Signal Specification

Signal Name	Underlying Metric / Formula	Weight	Target Hallucination Indicator
Token Probability	$S_{\text{prob}} = \exp(\text{avg_logprob})$	0.35	Low average token log-probability across response.
Predictive Entropy	$S_{\text{entropy}} = 1.0 - (H / \ln V)$	0.25	High Shannon entropy over token vocabulary distribution.
Semantic Similarity	$S_{\text{sem}} = (\cos(\theta) + 1.0) / 2.0$	0.25	Low embedding cosine similarity between prompt & response.
Self-Consistency	$S_{\text{sc}} = \text{Pairwise Cosine Similarity}$	0.15	High variance / low agreement across N sampled completions.

3. Directory Structure & Module Hierarchy

Module Path	Primary Responsibility	Key Classes / Functions
detector_agent/detector.py	Pipeline orchestrator for 4-signal aggregation	DetectorAgent
detector_agent/config.py	Pydantic configuration & signal weight validator	DetectorConfig, SignalWeights
detector_agent/model_manager.py	Singleton PyTorch LLM & SentenceTransformer loader	ModelManager
detector_agent/gate.py	Intelligent gating controller for Self-Consistency	SelfConsistencyGate
detector_agent/classifier.py	Heuristic rule-based prompt category classifier	PromptClassifier
detector_agent/datasets/	Hugging Face benchmark streaming infrastructure	BenchmarkExample, HaluEvalLoader, TruthfulQALoader
detector_agent/evaluation/	Classification, pairwise, ablation & dashboard suite	Evaluator, PairwiseEvaluator, AblationEvaluator, DashboardGenerator

4. Calibrated Thresholds & Risk Categorization

Hallucination Probability	Assigned Risk Level	Recommended Pipeline Action
Prob ≤ 0.40	LOW Risk	NextAction.ACCEPT (Bypass secondary verification)
0.40 < Prob < 0.55	MEDIUM Risk	NextAction.VERIFY (Triggers Gated Self-Consistency & Verifier)
Prob ≥ 0.55	HIGH Risk	NextAction.VERIFY (Escalate to Correction Agent)